

OSLAB PRACTICAL-3 PROGRAMS & OUTPUT

Date: 07-08-26

Program:

```
sainath@Sainath:~/OSLAB/practical_3$ cat program1.c
```

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid < 0)
    {
        printf("Fork Failed!\n");
        return 1;
    }

    else if(pid == 0)
    {
        // Child Process
        printf("\n----- CHILD PROCESS ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is Running...\n");
        sleep(5);

        printf("Child is Terminating...\n");
    }

    else
    {
        // Parent Process
        printf("\n----- PARENT PROCESS ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent is Waiting...\n");
        wait(NULL);

        printf("Child Process Completed.\n");
        printf("Parent Process Terminating...\n");
    }

    return 0;
}
```

OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_3$ ./program1
```

```
----- PARENT PROCESS -----
```

```
Parent PID : 697
```

```
Child PID  : 698
```

```
Parent is Waiting...
```

```
----- CHILD PROCESS -----
```

```
Child PID  : 698
```

```
Parent PID : 697
```

```
Child is Running...
```

```
Child is Terminating...
```

```
Child Process Completed.
```

```
Parent Process Terminating...
```

history:

```
238 cd OSLAB
239 mkdir practical_3
240 cd practical_3
241 gedit program1.c
242 gcc program1.c -o program1
243 ./program1
244 cat program1.c
245 ./program1
246 history
```